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Review

Jacaranda—An ethnopharmacological and phytochemical review

María Salomé Gachet, Wolfgang Schühly*

Institute of Pharmaceutical Sciences, Pharmacognosy, Karl-Franzens-University Graz, Universitätsplatz 4/I, 8010 Graz, Austria

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ABSTRACT

The genus *Jacaranda*, an important representative of the tribe Tecomeae in the family Bignoniaceae, is interesting from both biological and chemical perspectives. In this review, a contemporary summary of biological and pharmacological research on *Jacaranda* species will be presented and critically evaluated. Significant findings in the treatment of protozoa-caused diseases as well as of skin illnesses have been presented in ethnobotanical reports and recent studies were performed on crude extracts for certain *Jacaranda* species. Jacaranone, the most important constituent isolated is known to possess anti-cancer activity. Recently, high cutaneous toxicity together with moderate activity against leishmaniasis was described. Very few additional data are available on the biological activities and cytotoxicity of pure compounds from *Jacaranda*.

Thirteen of the forty-nine distinguished species of *Jacaranda* have been reported in scientific literature as ethnobotanically used or phytochemically investigated. However, information about a chemical profile is available only for six species. The following article gives a critical assessment of the literature to date and aims to show that the pharmaceutical potential of this genus has been underestimated and deserves closer attention.

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* Corresponding author. Tel.: +43 316 380 5527; fax: +43 316 380 9860.
E-mail address: wolfgang.schuehly@uni-graz.at (W. Schühly).

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1. Introduction

Jacaranda is a member of Bignoniaceae Juss., a dicot family that contains, according to respective taxonomic treatments, ca. 700–800 species grouped in 100–125 genera (Hegnauer, 1964; Mabberley, 1997). The Bignoniaceae is especially common in the tropics of South America and occurs in habitats consisting of mainly woody trees, shrubs, lianas and, rarely, herbaceous plants. Many woody representatives of Bignoniaceae are well known for their use in the timber industry (e.g. *Tabebuia*, *Cybistax*, *Paratecoma*, *Stereospermum*, *Parmentiera*, *Jacaranda*, *Catalpa*) or as ornamentals (e.g. *Jacaranda*, *Catalpa*, *Spathodea*). Some genera occur mainly as lianas such as *Pandorea* and *Macfadyena* (Willis, 1966; Parker, 2008).

Chemical constituents recognized in the family are naphthoquinones of the lapachol type, iridoid glucosides, alkaloids, flavones, triterpenes, polyphenols, tannins and seed oils (Hegnauer, 1964; Hegnauer, 1989). Lapachol (**35**) (Fig. 1) and some of its derivatives (α- and β-lapachone), recognized from the lapacho tree *Tabebuia avellanedae* (Hussain et al., 2007; Bentle et al., 2006), are found mainly in the heartwood of some bignoniaceous plants, as well as in a few species belonging to the families Verbenaceae, Proteaceae, Leguminosae, Sapotaceae, Malvaceae and Scrophulariaceae (Hussain et al., 2007; Budavari, 1996). Lapachol and β-lapachone have many pharmacological activities including anti-inflammatory, anti-microbial, anti-fungal, anti-viral, anti-parasitic, anti-protozoal and anti-cancer (Hussain et al., 2007).

Recently, Hussain et al. (2007) remarked on the promising biological activities of compounds containing a quinone moiety in their structure. These structures have been postulated to interfere with the bio-activities of *topoisomerase* enzymes, as well to induce reactive oxygen species, which cause cellular damage (Hussain et al., 2007).

Current research efforts focus on understanding the mode of action of β-lapachone. This compound has potential in cancer therapy, not only due to its cytotoxic activity (e.g. against cancer of the breast, colon, prostate, lung and pancreas), but also as a radiosensitizing agent that inhibits DNA repair in defective cells after irradiation, thus inducing apoptosis and preventing carcinogenesis (Hussain et al., 2007; Bentle et al., 2006; Bey et al., 2007; Krishnan and Bastow, 2001).

The genus *Jacaranda* is classified within the tribe Tecomeae together with the genus *Tabebuia* (Hegnauer, 1964; Heywood et al., 2007). Heywood et al. (2007) recognize this tribe as the only member of the Bignoniaceae that occurs in both the Old and the New World. From a chemotaxonomic point of view (Hegnauer, 1964), it is likely that *Jacaranda*, a close relative of *Tabebuia* may also contain lapachol-type compounds with promising activities.

1.1. *Jacaranda* Juss.

Jacaranda contains 49 species around the world that are native to Central and South America and the Caribbean. The majority of *Jacaranda* are trees ranging from 1 to 45 meters tall, but they can also be found as shrubs and subshrubs. Only *Jacaranda simplicifolia* has been recognized as a xylopodial herb or unbranched subshrub. Out of the 49 *Jacaranda* species, 39 taxa are endemic to Brazil. According to Gentry (1992), several of these endemic Brazilian species have been reported only from one location (e.g. *Jacaranda bullata*, *Jacaranda egleri*, *Jacaranda intricata*, *Jacaranda morii*, *Jacaranda paucifoliolata*, *Jacaranda praetermissa*), whereas *Jacaranda rugosa* is represented by only one herbarium voucher.

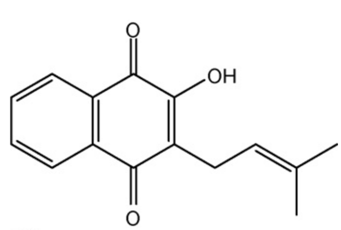

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Fig. 1. Molecular formula lapachol (Hussain et al., 2007).

Due to the rapid rate of world-wide deforestation in the tropics, which reached approximately 15% from 1990 to 2000 (FAO, 2001), the necessity of reviewing the current status of *Jacaranda* species is obvious. The species and subspecies of *Jacaranda*, together with their geographical locations, are presented in Table 1 .

2. Botanical characterizations

Jacaranda species may be identified by the disposition of the leaves, the type of inflorescence and the characteristics of the fruits, which according to Willis (1966), Gentry (1977, 1992) and Pennington et al. (2004) are

Leaves: Opposite, bipinnate and rarely pinnate or simple.
Inflorescence: Terminal, axillary or cauliflorous, paniculate, containing few to many flowers with a calyx campanulate to cupular. The corolla is blue to red, rarely white, usually 5-denticulate, 5-lobed, stamens with 1 or 2 thecae often 1; ovary 2-locular; pollen grains 3-colpate.
Fruits: Oblong capsule flattened perpendicular to the septum, the valves variously glabrous or lepidote; seeds numerous, thin, winged; the wings membranaceous, hyaline or brownish.

In general, *Jacaranda* occurs commonly as pioneer tree. It is light-demanding and colonize through wind-dispersed seeds where openings or gaps have been created by tree-fall. The genus has also been studied extensively in order to understand evolutionary parameters and studies of ecological relationships within tropical forests (Baraloto et al., 2005; Jones and Hubbell, 2003, 2006).

3. Economical and ecological considerations

As the nomenclature used in timber industry does not match the botanical nomenclature, confusion exists regarding the physical properties and timber value of *Jacaranda* trees. Various sources refer to *Jacaranda* species as South American rosewood (Bärner, 1943; Kunkel, 1978), tulipwood (FAO, 2001) or palisander (Bärner, 1943). These trade names used in the timber industry refer to high-grade hardwood with special properties and appearances, which are used in the production of finished articles of high value such as furniture. However, the common names of rosewood, tulipwood and palisander are also used for species of the genus *Dalbergia* and *Machaerium* (both Fabaceae). An additional confusion occurs because several species, which possess these high-grade wood qualities are commonly known as jacarandá. Examples of this are *Dalbergia spruceana* and *Dalbergia nigra* (jacarandá), *Dalbergia cearensis* (jacarandá violeta or pau violeta) and *Dalbergia frutescens* (jacarandá rosea or pau rosa). In the case of *Machaerium*, the species *Machaerium lanatum* and *Machaerium acutifolium* are known as jacarandá do campo; *Machaerium firmum*, as jacarandá rosa; *Machaerium villosum*, as jacarandá pardo; and *Machaerium scleroxylon*, as jacarandá caviuna or caviuna (Hueck and Seibert, 1981). Furthermore, the common names of many species of *Jacaranda* differ from “jacarandá”. Gentry (1992) reported

Table 1
Species of Jacaranda identified according the Flora Neotropica (Gentry, 1992).

Species	Cu	Ba	Ha	RD	Be	Ho	N	CR	Pan	Co	V	Gu	Su	FG	E	P	Bo	Br	Par	A
Jacaranda acutifolia Humb. & Bonpl.																x				
Jacaranda arborea Urban	x																			
Jacaranda bracteata Bureau & Schumann																		x		
Jacaranda brasiliana (Lam.) Persoon																		x		
Jacaranda bullata A. Gentry																		x		
Jacaranda caerulea (L.) Juss.	x	x	x	x																
Jacaranda campinae A. Gentry & Morawetz																		x		
Jacaranda carajasensis A. Gentry																		x		
Jacaranda caroba (Vellozo) A. P. de Candolle																		x		
Jacaranda caucana Pittier																				
Jacaranda caucana Pittier subsp. *caucana*										x	x									
Jacaranda caucana Pittier subsp. *glabrata* A. Gentry											x									
Jacaranda caucana Pittier subsp. *sandwithiana* A. Gentry								x	x	x										
Jacaranda caucana Pittier subsp. *calycina* A. Gentry										x										
Jacaranda copaia (Aublet) D. Don																				
Jacaranda copaia (Aublet) D. Don subsp. *copaia*											x	x	x	x				x		
Jacaranda copaia (Aublet) D. Don subsp. *spectabilis* (Mart. ex DC.)					x	x	x	x	x	x	x	x	x	x	x	x	x	x		
Jacaranda cowellii Bistton & Wilson	x																			
Jacaranda crassifolia Morawetz																		x		
Jacaranda cuspidifolia Martius ex. DC.																	x	x	x	x
Jacaranda decurrens Cham.																		x	x	
Jacaranda duckei Vattimo														x				x		
Jacaranda egleri Sandw.																		x		
Jacaranda ekmanii Alain			x	x																
Jacaranda glabra (DC.) Bureau & K. Schumann										x					x	x	x	x		
Jacaranda grandifoliolata A. Gentry																		x		
Jacaranda hesperia Dugand										x										
Jacaranda intricata A. Gentry & W. Morawetz																		x		
Jacaranda irwinii A. Gentry																		x		
Jacaranda jasminoides (Thunb.) Sandw.																		x		
Jacaranda macrantha Cham.																		x		
Jacaranda macrocarpa Bureau & K. Schum.										x						x		x		

Table 1 (*Continued*)

Jacaranda micrantha Cham.																		x	x	x
Jacaranda microcalyx A. Gentry																		x		
Jacaranda mimosifolia D. Don																	x			x
Jacaranda montana Morawetz																		x		
Jacaranda morii A. Gentry																		x		
Jacaranda mutabilis Hassler																		x	x	
Jacaranda obovata Cham.																		x		
Jacaranda obtusifolia Humboldt & Bonpland																				
Jacaranda obtusifolia Humb. & Bonpl. subsp. *obtusifolia*										x	x				x	x	x	x		
Jacaranda obtusifolia Humb.& Bonpl. subsp. *rhombifolia*											x	x	x	x				x		
Jacaranda orinocensis Sandw.											x									
Jacaranda oxyphylla Cham.																		x		
Jacaranda paucifoliolata Martius ex DC.																		x		
Jacaranda poitaei Urban			x	x																
Jacaranda praetermissa Sandw.																		x		
Jacaranda puberula Cham.																		x		x
Jacaranda pulcherrima Morawetz																		x		
Jacaranda racemosa Cham.																		x		
Jacaranda rufa Manso																	x	x		
Jacaranda rugosa A. Gentry																		x		
Jacaranda selleana Urban			x	x																
Jacaranda simplicifolia K. Schumann																		x		
Jacaranda sparrei A. Gentry															x	x				
Jacaranda subalpina Morawetz																		x		
Jacaranda ulei Bureau & K. Schumann																		x		

Abbreviations: Cu, Cuba; Ba: Bahamas; Ha, Haiti; RD, Dominican Republic; Be, Belice; Ho, Honduras; N, Nicaragua; CR, Costa Rica; Pan, Panama; Co, Colombia; V, Venezuela; Gu, Guyana; Su, Surinam; FG, French Guiana; E, Ecuador; P, Peru; Bo, Bolivia; Br, Brazil; Par, Paraguay; A, Argentina.

the common names of 19 of the 49 species of *Jacaranda*, and only *Jacaranda caucana* ssp. *caucana*, *Jacaranda copaia* ssp. *spectabilis* and *Jacaranda mimosifolia* are recognized as jacarandá.

The Tropical Timber Atlas of Latin America identifies the species of *Jacaranda* with the pilot name para-para (Chichignoud et al., 1990) and lists many other names used in different Latin American countries. In Table 2, the common names reported for 21 *Jacaranda* species, as well as their trade names, is presented.

According to Gentry (1992), the common name para-para is known only for the two subspecies of *Jacaranda copaia* and a very similar name pará-parai-mi is employed for *Jacaranda decurrens*. The name para-para is only recognized in Brazil, suggesting that Brazil might be the main producer of this timber. Additionally, *Jacaranda copaia*, considered a fast growing timber tree (Smith et al., 2004), is used in the production of paper pulp (Mabberley, 1997; Correa and Bernal, 1989) and table boards for construction (Correa and Bernal, 1989; Parker, 2008).

Butterfield has reported that *Jacaranda* species are among the best-performing species for commercial and plantation management in Costa Rica (Butterfield, 1994) and therefore they could represent a good alternative for use in reforestation projects in South and Central America.

4. Ethnobotanical studies

The traditional uses of *Jacaranda* species are referred to in many folkloric and ethnobotanical studies performed in certain areas of South America, where plants are still the primary source of medicine and where the species are native. All the studies reported in this section rely on interviews with native and non-native informants about the uses of plants against specific diseases.

Table 2
Common names reported for *Jacaranda* species.

Species	Common name
Jacaranda acutifolia	Yaravisco, paravisco; *in Bolivia* tarco[b]; *in Ecuador*: arabisco[b,e]
Jacaranda arborea	Abey
Jacaranda brasiliana	Castello do cavallo, caroba
Jacaranda caerulea	*In Bahamas*: boxwood, cancer tree, what-o'clock; *in Cuba*: abbey macho, framboyan azul, abey
Jacaranda caroba	Caroba, caroba muuda, carobinha[g], camboatá[g], camboatá-pequeno[g], cambotè[g], caroba-do-campo[g], caroba-miúda[g], carobado-carrasco[g]
Jacaranda caucana ssp. *caucana*	Gualanday, bayeto, jacaranda; *in Colombia*: acacia[b], aceituno[b], caballito[b], caro[b], cornique[b], gualanday[b], guayacán[b], palo de buba[b], piñón de oreja[b]
Jacaranda copaia ssp. *copaia*	*In Surinam*: koepaia, koepaja (Kar.), njamoesere (Kar.), goebai, goebaja, foete-i (Araw), jaifi (Bush Negro); *in Fr. Guiana*: jasi man boon (Ndjuka); *in Brazil*: marupauba, parapara, para-para, para'y (Ka'apor), *in Ecuador*: kupall yura (Kichwa)[e], qquepapajin (A'ingae)[e], hua'-hue (Pai coca)[e], kebamontowe (Wao tededo)[e], kuíship (Shuar chicham)[e]
Jacaranda copaia ssp. *spectabilis*	*In Panama*: cigarrillo, siete cueros, palo de buba[b], sule-grie (Guaymi); *in Colombia*: mallirokai (Witoto), canalete[b], caco castañete[b], chebin[b], chingale[b], escobillo[b], gallinazo[b], guaabandraño[b], guabillo[b], gualanday[b], guamoamu[b], madura plátano[b], papelillo[b], pavito[b], pinguasí, quitasol[b], sople[b], vainillo[b]; *in Venezuela*: girasol[a,b], flor azul[a,b], wei-oima-yek[a,b], palo azul[a,b], uai-cuima-yek[a,b], puti[a,b], pata de garza[a,b]; *in Fr. Guiana*: coupaya, copa la bois blanc, jaifi (Sar.), taki-taki (fenti), basa gobaja (Ndjuka), jasi man boon (Ndjuka); *in Ecuador*: gualandano, jacaranda, copa (Kichwa), quepapajin (Cofán), kuiship (Shuar), huilisha, (Palomino) gualanday[a,b], gualandray[e], qquepapajin (A'ingae)[e], guabandraño[b]; *in Peru*: huamansamana, huamanzamana, marupauba[b], ishtapi, cakaska (Jívaro); *in Brazil*: caroba, curoba, caruba, marupauba, parapara, para-para, para'y (Maranhão: Ka'apor)
Jacaranda decurrens	Pará parai mi, carobinha[f]
Jacaranda duckei	Paparauba de rato, caroba
Jacaranda glabra	*In Colombia*: gualanday[d]; *in Ecuador*: arabisca[e], kupall[e], kupall kaspi[e], kupall panka[e], kupall yura[e], ataira kaspi (Kichwa)[e], aya on'dui (A'ingae)[e]
Jacaranda jasminoides	Carobo, carobo muida
Jacaranda macrantha	Carabobinho, caroba
Jacaranda macrocarpa	Huamansamana
Jacaranda micrantha	Caroba, caroba do mato
Jacaranda mimosifolia	*In Argentina*: tarco, jacaranda[a,c], *in Ecuador*: arabisco[e], jacarandá[e], jaranda[e]
Jacaranda montana	Caroba
Jacaranda obovata	Carobinha
Jacaranda obtusifolia ssp. *obtusifolia*	*In Venezuela*: clavellina montanesa, clavellina, clavellino, guarupa, rabo de iguana, uadam-kayu, san jose, casabe, uada camayu, patico; in Colombia: gualanday[d]
Jacaranda obtusifolia ssp. *rhombifolia*	*In Venezuela*: clavellina montanesa, clavellina, clavellino, guarupa, rabo de iguana, uadam-kayu, san jose, casabe, uada camayu, patico; *in Surinam*: alasoeabo, itoeli walabang, koroballi (Arow.), alieskri, apakanipio, koepajarang, paloewe (Kar.), diamalieki, diomoliki, achewood, gobo-gobo wiwirie, goebar, kandra hoedoe, malokopesie, malimali
Jacaranda poitaei	Avay
Jacaranda puberula	Caroba, carobinha
Jacaranda rufa	Caroba do campo, perobinha, carobao
Jacaranda spp.	*In America Central*: jacaranda, palo de buba, samarapa, cola de zorro, tambor de montaña, gualanday; *in Argentina*: caroba, jacaranda, tarco; *in Bolivia*: tinto blanco; *in Brazil*: caroba, caroba manaca, caroba do mato, carauba, para-para, marupa falso; *in Colombia*: gualanday, chingale, pavito; *in Ecuador*: arabisco, kuiship; *in Guyana*: futui; *in Fr. Guiana*: copaia, yachimambo, bois pian; *in Peru*: chicharra, caspi, ishtapi; *in Surinam*: goebaja, foeti; *in Venezuela*: gualanday

Information: The main source of information is Flora Neotropica (Gentry, 1992) [a] and if not specified otherwise, the common names presented stem from this source. Flora Neotropica source [a] appears in the text only when a second source of information quotes the same common name. The names marked with [b] correspond to the source Vegetationskarte von Südamerika (Hueck and Seibert, 1981). The names marked with [c] correspond to the source Especies Vegetales Promisorias de los países del Convenio Andrés Bello (Correa and Bernal, 1989). The names marked with [d] correspond to Plantas útiles de Colombia (Pérez-Arbeláez, 1990). The names marked with [e] correspond to the source Plantas Útiles del Ecuador (De la Torre et al., 2007). The names marked with [f] correspond to the source Plantas Medicinais do Cerrado de Botucatu (Maroni et al., 2006). The names marked with [g] correspond to the source Plantas medicinais na Amazônia e na Mata Atlântica (Di Stasi and Hiruma-Lima, 2002). The last row of the chart catalogued as *Jacaranda* spp. contains the trade names of *Jacaranda* species which are reported in the book Tropical Timber Atlas of Latin America (Chichignoud et al., 1990).

4.1. *Jacaranda acutifolia*

The bark of *Jacaranda acutifolia* has been used in the treatment of wounds and dermatitis. Astringent and diuretic properties have also been assigned to the bark extracts (Roth and Lindorf, 2002). For this species, Correa and Bernal (1989) mention uses to treat wounds as antiseptic and to improve cicatrization. The treatment of wounds is carried out using a fresh decoction of the leaves (fresh or dry), or dried and powdered leaves are applied to the wounds directly. *Jacaranda acutifolia* has been attributed with properties to treat syphilis and diseases related to urinary tract problems. The ground bark is used as a decoction against venereal diseases or as ethanolic maceration, along with a small amount of *Cordia alliodora* (ajo de montaña, Boraginaceae), against rheumatism and sciatica (Correa and Bernal, 1989).

4.2. *Jacaranda caerulea*

The leafy branches of *Jacaranda caerulea* are used in Camaguey as decoction for bathing eczema and pimples. As well in Bahamas, the decoction of the parched leaves is used to treat skin cancer and other skin afflictions (Morten, 1981).

4.3. *Jacaranda caroba*

Jacaranda caroba has been used as a bitter or astringent, as diuretic and against syphilis. In some regions of Brazil, the leaves are applied externally during baths in the treatment of infections and infusions are taken internally to treat syphilis and as a cathartic remedy. The application of the macerated leaves in alcohol (cachaça) in cicatrization and internal ingestion to treat ulcers is

Table 3
Ethnobotanical uses recognized for *Jacaranda* species.

Species	L	SP	RD	GD	RC	RH	S	VD	UTA	BP	AD	F	GW	MR	FP/SB
Jacaranda acutifolia		x				x	x	x	x		x				
Jacaranda caerulea		x													
Jacaranda caroba		x						x		x	x				
Jacaranda caucana		x	x			x		x		x					
Jacaranda copaia	x	x	x	x	x	x		x					x	x	x
Jacaranda cuspidifolia	x														
Jacaranda decurrens		x				x		x		x					
Jacaranda glabra	x	x						x							
Jacaranda hesperia	x														
Jacaranda mimosifolia								x		x					
Jacaranda obtusifolia		x						x							
Jacaranda puberula												x			

L, leishmaniasis; SP, skin problems; RD, respiratory diseases; GD, gastrointestinal diseases; RC, ritualistic cleansing; RH, rheumatism; S, sciatica; VD, venereal diseases; UTA, urinary tract affections; BP, blood purification; AD, astringent and diuretic; F, frostbite; GW, general weakness and fever; MR, mosquito repellent; FP/SB, fish poison/snake bites.

widely practiced (Di Stasi and Hiruma-Lima, 2002; Botion et al., 2005).

As well in Brazil, the habitants of Vale do Ribeira, São Paulo, employed this species as depurative, tonic, to combat syphilis, wounds and ulcers (Di Stasi and Hiruma-Lima, 2002).

4.4. *Jacaranda caucana*

Gentry (1992) reported that an infusion of the leaves and bark of *Jacaranda caucana* ssp. *caucana* is used in the treatment of venereal diseases. Correa and Bernal (1989) also recognized the use of the leaves of this species in the treatment of syphilis and as blood purifier. The warm decoction of the leaves or bark is applied externally on affected parts to treat ulcerations and phlegmons or taken orally. Furthermore, dry powdered leaves are applied onto ulcerations and are believed to be disinfectant.

Weniger et al. (2001) cited that *Jacaranda caucana* and other species of the same genus are used in Colombia against rheumatism, colds and skin diseases.

4.5. *Jacaranda copaia*

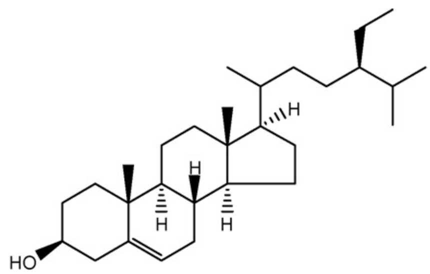
With regard to *Jacaranda copaia*, studies performed in the Amazon region described two different uses, external and internal. The external use relates to the treatment of skin infections through application of the sap of the bark and leaves. Additionally, the Andoque Indians in the Colombian Amazon prepare a decoction of the crushed leaves until it reaches a honey-like consistency for cicatrization and apply this remedy topically to promote healing (Correa and Bernal, 1989; Evans-Schultes and Raffauf, 1990). The internal uses include the preparation of a hot decoction of the shredded bark, taken for the treatment of strong colds or pneumonia (Evans-Schultes and Raffauf, 1990). The use of *Jacaranda copaia* in the treatment of skin-related diseases is also reported from the Wao and Shuar Indians of the Ecuadorian-Amazon region. The leaves are applied onto wounds externally, either after being boiled in water or dry and powdered (De la Torre et al., 2007). The tubercles of *Jacaranda copaia* are used by the traditional healers in the Brazilian Amazon and are used for the treatment of gastrointestinal disturbances (Rodrigues, 2006). A cold infusion prepared from the chopped roots is used in low doses (one to two spoonfuls) to treat diarrhea, but when taken at higher concentrations, it induces vomiting (Correa and Bernal, 1989). Traditional healers use *Jacaranda copaia* to cleanse the stomach before beginning a specific diet as part of the preparation for healing practices (Correa and Bernal, 1989). Furthermore, the bark of *Jacaranda copaia* has been used for the treatment of leishmaniasis in South America (Roth and Lindorf, 2002) and, in particular, this practice is performed among the populations inhabiting the tableland of the Guiana's (Sauvain et al., 1993). Roth and Lindorf (2002) reported also the use of *Jacaranda copaia* in Venezuela as vulnerary, cicatrizing, to prevent infections and against cancer.

Jacaranda copaia is only used by the Waimiri Atroari Indians for construction and fuel however, Milliken et al. (1992) also presented additional ethnobotanical information on the use of this species in the Americas. In Bolivia, the Chácobo Indians prepare a decoction of the leaves which is drunk to treat rheumatism. The Tiriyó of northern Brazil use the infusion of the leaves of *Jacaranda copaia* in a bath to heal general weakness and fever. The Wayãpi Indians of the French Guinea and in the people from the Alter do Chão, Pará region in Brazil, used the burned leaves as black-fly and mosquito repellent, respectively. In the French Guiana, the bark of young trees is used to treat syphilis as well as to prepare a purgative. Additionally, the use of the leaf sap in the treatment of leishmaniasis has also been mentioned by the Negroes in the French Guiana. The Jívaros of Perú applied the crushed leaves topically to treat skin infections

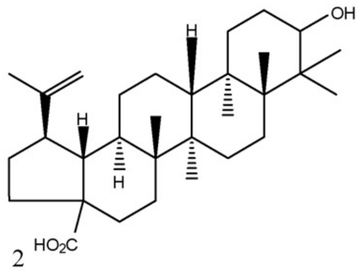
Table 4
Resume of activity reported on the crude extracts of *Jacaranda* species.

	P.f.	*L.*	*T.c.*	*P.e.*	*A.n.*	*C.v.*	*G.t.*	*B.t.*	*B.c.*	*B.s.*	*E.c.*	*S.a.*	*C.a.*	HP	DP	NF	P
Jacaranda caroba															A		
Jacaranda caucana	A	NA	NA														
Jacaranda copaia																A	NA
Jacaranda cuspidifolia		NA	NA														
Jacaranda filicifolia				NA	NA	A	A	A									
Jacaranda glabra										NA		NA	NA				
Jacaranda mimosifolia									A		A	A		A			
Jacaranda puberula		A															

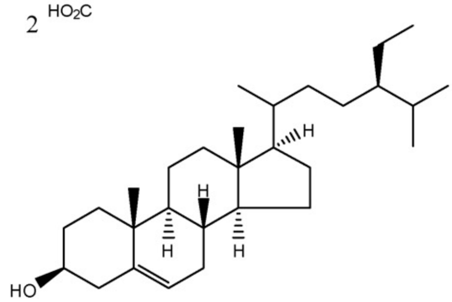
A, active (when any kind of activity was reported); NA, not active (when no activity was stated). *Plasmodium falciparum*: *P.f.*, *Leishmania* spp.: *L.*, *Trypanosoma cruzi*: *T.c.*, *Penicillium expansum*: *P.e.*, *Aspergillus niger*: *A.n.*, *Coriolus versicolor*: *C.v.*, *Gloeophyllum trabeum*: *G.t.*, *Bostryodiplodia theobromea*: *B.t.*, *Bacillus cereus*: *B.c.*, *Bacillus subtilis*: *B.s.*, *Escherichia coli*: *E.c.*, *Staphylococcus aureus*: *S.a.*, *Candida albicans*: *C.a.*, hypotensive properties: HP, digestive properties: DP, NF-κB: NF, Proteases: PR.



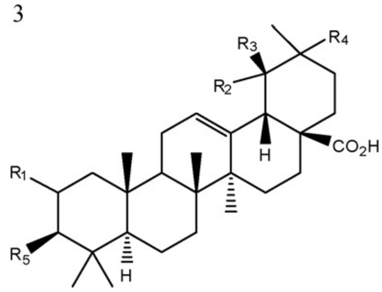
1 β-Sitosterol[a,b,c]

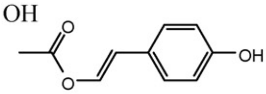


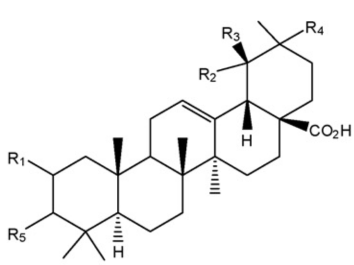
2 Betulinic acid[d]



3 Lupenone[c]

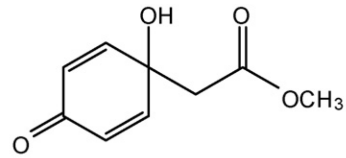


	R_1	R_2	R_3	R_4	R_5	
4	H	H	Me	H	OH	Ursolic acid[a,b,e,f]
5	OH	H	Me	H	OH	2α-hydroxyursolic acid[a]
6	H	H	H	Me	OH	Oleanolic acid[c]
7	OH	H	Me	H		Jacoumaric acid[a]

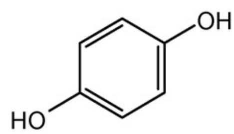


	R_1	R_2	R_3	R_4	R_5	
8	OH	H	Me	H	OH	2α,3α-Dihydroxyurs-12-en-28-oic acid[a,b]
9	OH	OH	Me	H	OH	Jacarandic acid[a]

Fig. 2. Triterpenes isolated from *Jacaranda* species.
[a]Isolated from *Jacaranda caucana* (Ogura et al., 1977a). [b]Isolated from the stem of *Jacaranda filicifolia* (Ali and Houghton, 1999). [c]Isolated from the root bark of *Jacaranda mimosifolia* (Prakash and Garg, 1980). [d]Isolated from the stem-bark of *Jacaranda caucana* (Ogura et al., 1977b). [e]Isolated from the leaves of *Jacaranda copaia* (Sauvain et al., 1993). [f]Isolated from the epicuticular wax of *Jacaranda decurrens* (Varanda et al., 1992).



10 Jacaranone[a,b]



11 Hydroquinone[c]

Fig. 3. Quinones isolated from *Jacaranda* species. [a]Isolated from the twigs-leaves (Ogura et al., 1976) and from the stem-bark (Ogura et al., 1977a) of *Jacaranda caucana*. [b]Isolated from the leaves of *Jacaranda copaia* (Sauvain et al., 1993). [c]Isolated from the leaves of *Jacaranda mimosifolia*.

and for the same purpose, the Vaupés River Indians in Colombia use the sap of the bark. The used of this species in the Guiana's have been reported to be employed as fish poison (Milliken et al., 1992).

Interestingly, already Hager and Geissler (1882) reported the use of the leaves of *Jacaranda copaia* (referred to as *Jacaranda procerae*) in the treatment of syphilis, skin affections and snake bites.

4.6. *Jacaranda cuspidifolia*

In Bolivia, the leaves of *Jacaranda cuspidifolia* are used in the treatment of leishmaniasis among the Chinane Indians and Colonos (Fournet et al., 1994).

4.7. *Jacaranda decurrens*

In Brazil, the infusion and decoction of the leaves and bark of *Jacaranda deurrens* are employed as depurative and to treat cutaneous disturbances and wounds. It has been cited that a certain preparation called "garrafada", made from the wine of the roots and leaves of *Jacaranda deurrens*, is used against syphilis, rheumatism, inflammation and dermatological diseases. Furthermore, the infusion of the bark is employed to cure itching (Maroni et al., 2006).

4.8. *Jacaranda glabra*

The Tacana Indians in Bolivia apply the mashed leaves of *Jacaranda glabra* topically to treat leishmaniasis (Bourdy et al., 2000). Similarly, the Kichwas of the Ecuadorian Amazon use the decoction of branches and leaves of *Jacaranda glabra* as a wash in leishmaniasis. The Amazon Kichwas in Ecuador also use *Jacaranda glabra* in the form of an infusion of the dry leaves to treat other skin problems such as infected pin polls, scabies, sores and fungal infection (De la Torre et al., 2007).

4.9. *Jacaranda hesperia*

Afro-American informants from the Chocó region in Colombia have reported to use *Jacaranda hesperia* in the treatment of leishmaniasis (Vázquez et al., 1991).

4.10. *Jacaranda mimosifolia*

A decoction of the bark of *Jacaranda mimosifolia* is used in Ecuador to treat venereal diseases and to purify the blood (Acosta-Solís, 1992).

4.11. *Jacaranda obtusifolia*

In Venezuela and Guyana, Roth and Lindorf (2002) reported the use of the bark of *Jacaranda obtusifolia* in order to promote wound healing and as disinfectant. Pérez-Arbeláez (1990) recognized the use of the leaves of certain species of *Jacaranda* in Colombia as a popular remedy to treat syphilis. Gualanday, one of the common names used for *Jacaranda* in Colombia, may refer to *Jacaranda obtusifolia, Jacaranda glabra* and/or *Jacaranda caucana.*

4.12. *Jacaranda puberula*

An interesting use has been documented for the leaves of *Jacaranda puberula* among the Xokleng Indians of Terra Indígena Ibarama, living in southern Brazil. The water decoction of the leaves (three full handfuls boiled in one liter of water) is added to baths for the treatment of frostbite (Sens, 2002).

In Table 3, the ethnobotanical information reported for *Jacaranda* species is summarized.

5. Biological activity of crude extracts

In most biological studies performed, *Jacaranda* extracts were identified as active against specific biological targets during large-scale screenings of multiple plant extracts.

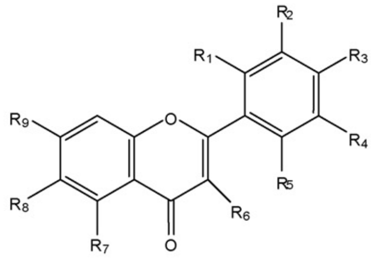
The methanolic extract of the leaves of *Jacaranda caucana* was tested *in vitro* against two strains of *Plasmodium falciparum*. For the chloroquine-sensitive strains, the reported IC_{50} was 14 μg/mL at 72 h, while for the chloroquine-resistant IC_{50} was 4.6 μg/mL. Cytotoxicity, evaluated using human promonocytic U-937 cells, was low (LD_{50} 280 μg/mL). Anti-leishmanial and anti-trypanosomal activities were not found (Weniger et al., 2001).

The methanol extract of the leaves of *Jacaranda puberula* was recently shown to have a certain anti-leishmanial activity *in vitro* against *Leishmania amazonensis* promastigotes and amastigotes. The activity against promastigotes was moderate (IC_{50}: 88 μg/mL at 72 h), but against amastigotes was weak (IC_{50}: 359 μg/mL at 24 h). In both cases, at higher concentrations toxicity was observed (Passero et al., 2007).

The petroleum ether, ethyl acetate and ethanol/water extracts of the leaves of *Jacaranda cuspidifolia*, targeted due to their use in the treatment of leishmaniasis in Bolivia, were found to be inactive *in vitro* against promastigotes of *Leishmania* species and *Trypanosoma cruzi* epimastigotes at 100–5 μg/mL (Fournet et al., 1994).

A test of the dichloromethane extract of *Jacaranda filicifolia* stem bark against *Penicillium expansum* and *Aspergillus niger* using the agar diffusion method (100 μg application) did not show positive results. In contrast, moderate antifungal activity against *Coriolus versicolor*, *Gloeophyllum trabeum* and *Bostryodiplodia theobromae* at a concentration of 1.7 mg/mL was observed (Ali et al., 1998). Chichignoud et al. (1990) note that many *Jacaranda* species seem to be poorly resistant against insects and termites, e.g. *Jacaranda copaia* is not resistant to insects or decay (Parker, 2008).

Antimicrobial activity of the hexane, ethanol and water extracts of the leaves of *Jacaranda mimosifolia* against *Bacillus cereus*, *Escherichia coli* and *Staphylococcus aureus* using the agar diffusion method at a concentration of 0.6 μg have been reported. Results showed an activity against *Bacillus cereus* (minimal inhibitory concentrations (MIC) of 2.9, 3.3 and 4.0 μg/mL for the hexane, ethanol and water extracts, respectively), for the hexane and water extracts against *Escherichia coli* (MIC of 2.3 and 14.4 μg/mL) and for the ethanol extract against *Staphylococcus aureus* (MIC of 2.8 μg/mL) (Rojas et al., 2006).



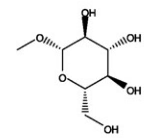
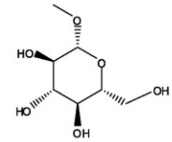
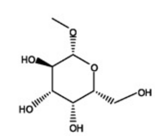
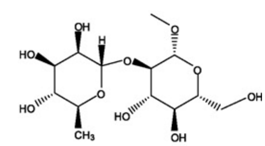
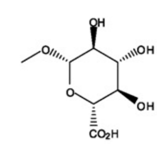
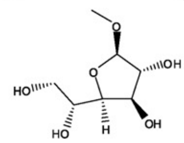
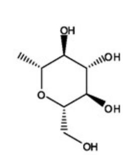
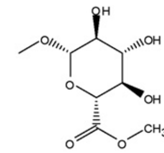
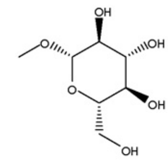
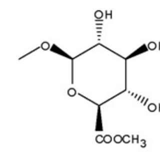
	R_1	R_2	R_3	R_4	R_5	R_6	R_7	R_8	R_9
12	H	H	OH	OH	H	H	OH	H	OH
13	H	H	OH	OH	H	H	OH	OH	
14	H	OH	OH	H	H	OH	H	OH	
15	H	OH	OH	H	H	OH	H	OH	
16	OH	OH	OH	H	H	OH	H	OH	
17	H	H	OH	H	H	H	OH	OH	OH
18	H	H	OH	H	H	H	OH	OH	
19	H	OH	OH	H	H		OH	H	OH
20	H	OH	H	H	H	H	OH		OH
21	H	H	OH	H	H	H	OH	H	

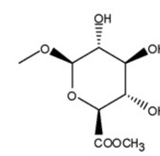
Fig. 4. Flavonoids isolated from *Jacaranda* species.
Luteolin[a] **12**, 6-hydroxyluteolin 7-*O*-glucoside[a] **13**, quercetin-3-*O*-glucoside[a] **14**, quercetin-3-*O*-galactoside[a] **15**. 7,2′,3′,4′-tetrahydroxyflavone 3-*O*-neohesperidoside[b] **16**, scutellarein[c] **17**, scutellarein 7-glucuronide[d] **18**, isoquercitrin[e] **19**, isovitexin[e] **20**, apigenin 7-*O*-β-D-glucoronopyranoside[e] **21**, luteolin 7-*O*-β-D-glucopyranoside[e] **22**, scutellarein 7-*O*-β-D-glucoronopyranoside methyl ester[e] **23**, apigenin 7-*O*-β-D-glucuronopyranoside methyl ester[e] **24**, luteolin 7-*O*-β-D-glucuronopyranoside methyl ester[e] **25**.
[a]Identified from the leaves of *Jacaranda decurrens* (Blatt et al., 1998). [b]Isolated from the bark of *Jacaranda acutifolia* (Ferguson and Lien, 1982). [c,d]Isolated from the leaves of *Jacaranda mimosifolia* (Sankara-Subramanian et al., 1972)[c], (Sankara-Subramanian et al., 1973)[d]. [e]Isolated from the leaves of *Jacaranda mimosifolia* (Moharram and Marzouk, 2007).

22	H	H	OH	OH	H	H	OH	H
23	H	H	OH	H	H	H	OH	OH
24	H	H	OH	H	H	H	OH	H
25	H	H	OH	OH	H	H	OH	H









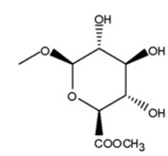
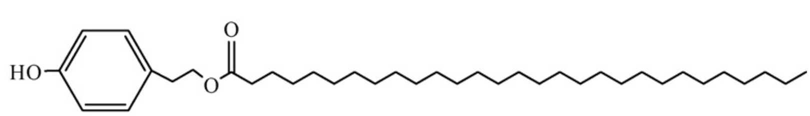


Fig. 4. (*Continued*)

The hypotensive properties of the methanol–water extract of the leaves of *Jacaranda mimosifolia* have been studied. *In vivo* experiments performed in male Wistar rats revealed hypothermic (IC_{50} of 100 mg/kg; LD_{50} of 655 mg/kg) and cardiovascular depressive (at 100–400 mg/kg) activities of the extract. The hypotensive properties reported *in vitro* (evaluated through analysis of contractile tension development) suggest a mode of action via a blockage of α-adrenergic receptors (Nicasio and Meckes, 2005).

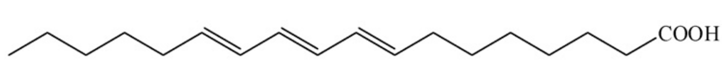
Antimicrobial studies using the agar disk diffusion method (200 μg application) of the hexane, dichloromethane and methanol extracts of the leaves of *Jacaranda glabra* indicated no activity against *Staphylococcus aureus*, *Bacillus subtilis* and *Candida albicans* (Gachet et al., 2007).

An ethanol extract of the aerial parts of *Jacaranda caroba* is one of four plant ingredients contained in the Brazilian phytopharmaceutical product Ierobina®, used in the treatment of dyspepsia. *In vivo*



26

2-(4-Hydroxyphenyl)ethyl 1-dodecyloctadecanoate (triacontanoic acid) [a]



27

8Z, 10 E, 12 Z-octadocatrienoic acid [b]

Fig. 5. Fatty acids isolated from *Jacaranda* species. [a]Isolated from the stem of *Jacaranda filicifolia* (Ali and Houghton, 1999). [b]Isolated from the seed oil of *Jacaranda mimosifolia* (Chisholm and Hopkins, 1962).

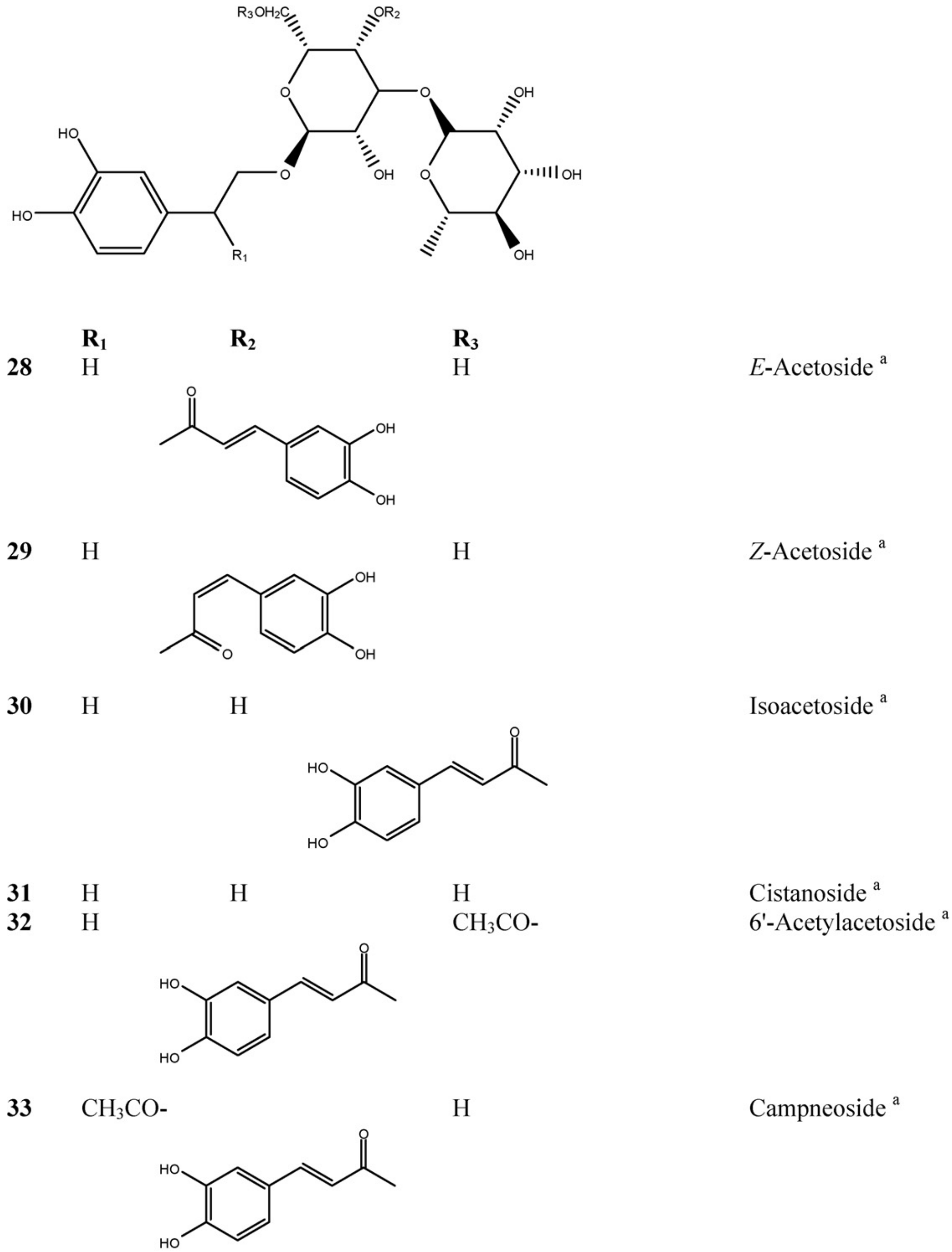
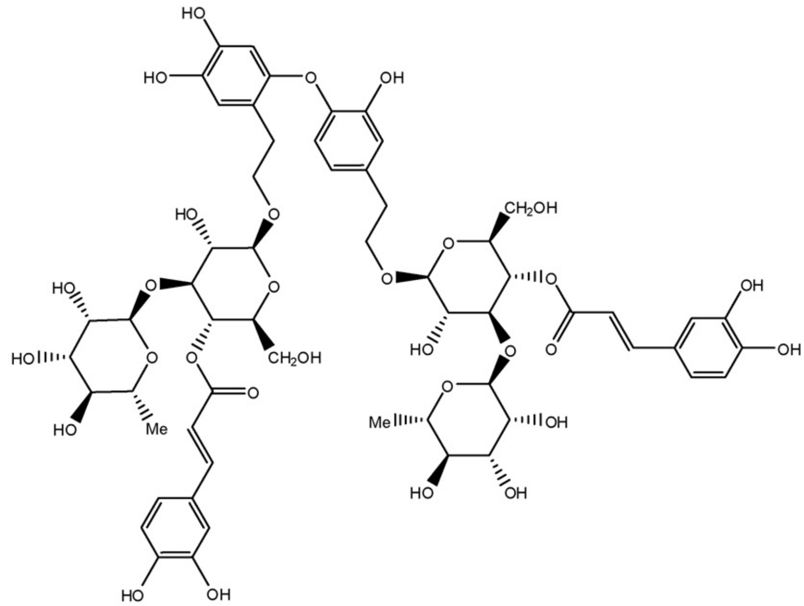


Fig. 6. Acetosides isolated from *Jacaranda* species. [a]Identified from the leaves of *Jacaranda mimosifolia* (Moharram and Marzouk, 2007).

studies in male Wistar rats, held on a high-fat diet, have revealed the modulating influence of the product on the triacylglycerol TAG levels (Botion et al., 2005).

Recently, two similar studies have investigated the anticancer properties of the ethanolic extract of *Jacaranda copaia* on different cell-lines together with the NF-κB effect in HeLa cells (young leaf) (Villasmil et al., 2006) and the activity against four proteases (young leaf and internal bark) (Taylor et al., 2006). Even though the antitumor activity (attributed to the leaf extracts) is known and immediately related to jacaranone and its toxicity (see below), the positive result obtained for the NF-κB inhibitions $IC_{50} = 10\,\mu g/mL$, which is not due to short-term toxicity on HeLa cells cannot be correlated (Villasmil et al., 2006). In regard to effect on the proteases assayed (LAP leucineaminopeptidase, elastase, CAT-D cathepsin-D and CAT-B) none of the extracts showed inhibition higher that 50% at a concentration of 100 μg/mL (Taylor et al., 2006).

A general overview of the reported activities of the extracts of the seven *Jacaranda* species presented in this section can be found in Table 4.



34
Jacraninoside A

Fig. 7. Novel phenylethanoid Isolated from the leaves of *Jacaranda mimosifolia* (Moharram and Marzouk, 2007).

6. Chemical constituents

Chemical studies on the constituents of *Jacaranda* have only been reported for six species: *Jacaranda acutifolia* (Ferguson and Lien, 1982), *Jacaranda caucana* (Ogura et al., 1976, 1977a,b), *Jacaranda copaia* (Sauvain et al., 1993), *Jacaranda decurrens* (Blatt et al., 1998; Varanda et al., 1992), *Jacaranda filicifolia* (Ali and Houghton, 1999) and *Jacaranda mimosifolia* (Chisholm and Hopkins, 1962; Sankara-Subramanian et al., 1972, 1973; Prakash and Garg, 1980; Moharram and Marzouk, 2007). The compounds have been identified as triterpenes (Fig. 2), quinones (Fig. 3), flavonoids (Fig. 4), fatty acids (Fig. 5), acetosides (Fig. 6), and, recently, a novel phenylethanoid dimmer (Fig. 7). In the respective figures, a full description of the phytochemistry of these species, the part of the plant investigated and the molecular structures are presented.

Jacaranone (**10**), obtained from the water extract of the twig-leaf and stem bark of *Jacaranda caucana*, showed important *in vivo* and *in vitro* anticancer activity against P-388 lymphocytic leukemia cells at 2 mg/kg with marginal cytotoxicity in Eagles 9KB carcinoma of the nasopharynx (Ogura et al., 1976, 1977a). Because of its high drug potential, jacaranone was patented for antitumor activity in 1977 (Farnsworth et al., 1977). Since then, jacaranone and other derivatives have been described from other species (Pérez-Castorena et al., 2001; Torres et al., 2000; Yvin et al., 1985) and its synthesis has also been published (Lasne et al., 1980; Parker and Andrade, 1979). The moderate anticancer activity of a naturally occurring jacaranone glycoside has also been determined (Tian et al., 2006).

Jacaranone was later also isolated from the leaves of *Jacaranda copaia* during a study of its anti-leishmanial activity. Tests with jacaranone at a concentration of 0.33 mM/kg daily for 5 days, or when applied as a single dose at 0.41 mM/kg against *Leishmania amazonensis in vivo* on BALB/c infected mice, showed strong cutaneous toxicity, an inflammatory reaction and low specificity when applied subcutaneously. Additionally, toxicity assays performed have led to an estimated LD_{50} value of jacaranone between 67 mg/kg (total survival) and 200 mg/kg (no survival) (Sauvain et al., 1993). Derivatives were prepared to establishing the anticancer activity of jacaranone-related naphthoquinones in a KB *in vitro* test system (Deutsch and Zalkow, 1982).

Moderate toxicity of jacaranone was demonstrated against many organisms. Jacaranone inhibits the growth of tobacco cutworm larvae (*Spodoptera litura*) and the root growth of some vegetables. It is toxic to brine shrimp (*Artemia salina*), induces metamorphosis in marine invertebrate larvae (*Pecten maximus*) (Yvin et al., 1985), has insecticidal properties and antimicrobial activity (Xu et al., 2003).

Toxicity studies performed by Xu et al. (2003) in male albino Swiss-Webster mice have associated the mode of action of jacaranone with neurotoxicity (LD_{50} = 150–200 mg/kg intra peritoneal i.p.) and also with glutathione GSH depletion in the liver.

In addition, 8 *Z*, 10 *E*, 12 *Z*-octadocatrienoic acid (**27**) found as the main component of the seed oil of *Jacaranda mimosifolia* (Chisholm and Hopkins, 1962) reported high cyclooxygenase activity (IC_{50} = 2.4 mM) (Nugteren and Christ-Hazelhof, 1987). However, not further information on regard to this compound is available. Interestingly, none of these two sources reported NMR data or detailed GC–MS information. The only detail information that characterized the structure is based on melting points, UV and IR analysis. Furthermore, it seems also to be confusion on the trivial name given by Nugteren and Christ-Hazelhof (1987). Jacarandic acid **9** is used to refer the 2α, 3α, 19α-trihydroxyurs-12-en-28-oic acid and not 8 *Z*, 10 *E*, 12 *Z*-octadocatrienoic acid **27**.

Recently, Moharram and Marzouk (2007) published information on a novel phenylethanoid dimer jacaraninoside (**34**).

7. Conclusions and outlook

Reports for biological activity of 12 of the 49 species of *Jacaranda* exist, but phytochemical investigations have been conducted on only six of them. Quinone-type compounds seem to be typical for the genus as they were found from several species. Among these compounds, jacaranone (**10**), is most remarkable and has received much scientific attention because of its anti-cancer potential. However, the high level of cutaneous toxicity (Sauvain et al., 1993) indicates that further development may be unrealistic.

Significant activity of the methanol extract of certain species against *Plasmodium* was observed, along with low toxicity (Weniger et al., 2001). Because of the low toxicity observed, it could be assumed that in this extract jacaranone was not present.

Information on uses related to venereal diseases is representative and frequent among the available ethnobotanical information. However, the pharmacological activities in regard to this treatment have not yet been evaluated.

Jacaranda is frequently used in the treatment of skin diseases. Due to the wide distribution and known wound healing properties, tannins and polyphenols are expected to be responsible for these activities, which has been demonstrated *in vivo* by Chaudhari and Mengi (2006). In the context of cutaneous leishmaniasis, Kolodziej and Kiderlen (2005) suggested that tannins may have anti-leishmanial activity and even immune modulatory effects *in vitro*. Several studies have been carried out to evaluate the pharmacological activity of *Jacaranda*, but an active principle could not yet be identified.

Ethnobotanical studies based on interviews clearly state a traditional use of *Jacaranda* in the treatment of leishmaniasis (Weniger et al., 2001; De la Torre et al., 2007; Fournet et al., 1994). In contrast, a similar investigation performed in the Peruvian Amazon documented 85 plant-taxa and did not report any *Jacaranda* species (Kvist et al., 2006). Evidently, the differentiation between the different types of skin diseases remains often vague. An important point to consider is that the interviewing process and the ability of the local people to distinguish the difference between leishmaniasis and other sores and wounds may influence the accuracy of the ethnobotanical information obtained. This is illustrated by two studies performed in Colombia, which revealed that the people who were interviewed do not have a clear conception of leishmaniasis (i.e. the mode of transmission, the vector and the life cycle) (Vázquez et al., 1991; Isaza et al., 1999). Therefore, it could be assumed that treatment of leishmaniasis implies only the treatment of obvious cutaneous problems, such as ulcers. The underlying disease, i.e. leishmaniasis, is not targeted.

These healing properties reported in ethnobotanic surveys could be also associated with activities of isolated compounds reported in *Jacaranda*. For example, oleanolic acid (**8**) has shows wound healing properties in tests performed *in vivo* (Moura-Letts et al., 2006; Rodríguez et al., 2003) as well as moderate anti-leishmanial activity reported *in vitro* (Torres-Santos et al., 2004). Luteolin (**12**) showed anti-leishmanial activity (Tasdemir et al., 2006) by inducing apoptosis-like death in promastigotes and amastigotes (Sen et al., 2006) *in vitro*.

Clearly, members of the genus *Jacaranda* possess significant pharmacological potential and promising activities of extracts in the context of ethnomedicinal knowledge, especially in the field of tropical diseases, skin problems and venereal illnesses and therefore promote a high degree of interest in further studies. Pharmacological studies should be carried out considering the traditional mode of preparation and application. Knowledge obtained from such studies could also enhance the efficacy of already existing ethnomedicinal uses and, consequently, support the cultural value of these species. The *Jacaranda* species described in this review do not appear limited in their availability and might serve as an important source of medicine among people living in tropical regions.

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